Fiscal Year:	FY 2026	**Task Last Updated:**	FY 09/03/2025
PI Name:	Furst, Eric Ph.D.		
Project Title:	Self-Assembled Microstructure Growth in Directing Fields		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	furst@udel.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	
Organization Name:	University of Delaware		
PI Address 1:	Chemical & Biomolecular Engineering		
PI Address 2:	150 Academy Street		
PI Web Page:			
City:	Newark	**State:**	DE
Zip Code:	19716	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2022 Physical Sciences NNH22ZDA001N-PSI E.8 Physical Sciences Informatics
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Hasan, Mohammad	**Contact Phone:**	216-977-7494
Contact Email:	Mohammad.M.Hasan@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2026 per J. McQuillen/GRC (Ed., 9/29/25). NOTE: End date changed to 09/30/2025 per J. McQuillen/GRC (Ed., 6/5/25).		
Key Personnel Changes/Previous PI:	No changes to report.		
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Task Description:	Colloids and nanoparticles self-assemble in external fields, a property that enables functional and smart materials like magnetorheological fluids, and potentially new ones in which structure controls the transport of heat, light, or chemical species. Paramagnetic colloids are responsive suspensions that self-assemble; in a strong, steady magnetic field, paramagnetic colloids form system-spanning, kinetically arrested networks analogous to a gel. From this state, it is possible to phase separate and condense the suspension by steadily toggling the external field on and off. During its evolution towards the equilibrium state, particles initially chain together to form a percolated network that coarsens diffusively. With time, the surface of the growing domains in the network become unstable. The amplitude of the waves eventually reaches a critical value, and the columns pinch off and condense into anisotropic droplet-like structures. I propose to use the InSPACE-1 and InSPACE-2 studies of paramagnetic colloid assembly in steady and toggled fields to advance the understanding of dissipative self-assembly, a generalization of processes in which ordered structures form far from equilibrium by continuously absorbing energy and dissipating it into the surroundings. Such active assembly processes offer promising routes to generate complex structures and circumvent arrest in undesirable metastable states. The InSPACE-1 Physical Sciences Informatics (PSI) data package contains unique experiments that provide structure in steady fields. These are an important basis for later experiments (InSPACE-2, InSPACE-4) that examined colloidal self-assembly in toggled fields. In the latter experiments, it was found that several factors govern the kinetics of phase separation of paramagnetic colloids and the steady-state structures that they form in toggled fields. These include the field strength, toggle frequency, and duty ratio. The suspension structures observed over many toggle cycles appear to be in agreement with those predicted by the theoretical and computational work of Sherman et al. [Z. M. Sherman, H. Rosenthal, J. W. Swan, Langmuir 2018, 34, 1029], including a dependence on the toggled field duty fraction. We will analyze the structure development of the InSPACE-1 and InSPACE-2 video microscopy data in the PSI system by developing characterization methods based on real- and Fourier-space analysis. The InSPACE-1 data is unique in that it provides views of the evolving microstructure with time, along the applied magnetic field in steady fields that are complementary to subsequent experiments in InSPACE-2. As a second aim, we will adapt open-source Brownian dynamics simulations that incorporate hydrodynamic interactions to model the InSPACE-1 and InSPACE-2 experiments. By comparing simulation results to the rich, but limited, video microscopy results of the microgravity experiments, we can develop an understanding of microstructure evolution during dissipative assembly that has so far been inaccessible. Similarly, the experimental data represents important tests for computational modeling. The proposed work is significant because it will substantially advance our ability to model dissipative self-assembly and harness it to create new functional and possibly responsive micro- and nano-structured materials, some of which will be realized in future ground-based experiments. In this project, we expect to create or substantially advance more general numerical, analytical, and simulation tools using the InSPACE-1 and InSPACE-2 data packages.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our results are important experimental tests for models of dissipative self-assembly -- processes in which ordered structures form far from equilibrium by continuously absorbing energy and dissipating it into their surroundings. Such active assembly processes offer promising methods to generate complex structures and circumvent arrest in undesirable metastable states. The result will be new functional and smart materials like magnetorheological fluids and potentially others in which structure controls the transport of heat, light, or chemical species.
Task Progress:	During this reporting period, our team advanced the analysis of unique microgravity data from the InSPACE experiments, applying modern image-processing and modeling methods that were unavailable at the time the experiments were conducted. These studies examine the self-assembly of polarizable colloidal particles in magnetic fields – both steady and toggled – and offer rare insight into the physical mechanisms driving the formation of anisotropic structures in microgravity. Our work is generating new understanding of dissipative self-assembly, with results disseminated through national and international conferences, invited talks, and multiple manuscripts. One paper has been published, two are under review, and three more are in preparation. The project has also supported the training of a graduate student in advanced modeling, data analysis, and scientific communication. The central goal is to analyze and interpret InSPACE-1 and InSPACE-2 data sets, archived on NASA's Physical Sciences Informatics platform, to test and refine theories of dissipative self-assembly. InSPACE-1 data – never before analyzed for the archival literature – offers a rare steady-field microgravity dataset for comparison with recent experiments (including InSPACE-4, performed in 2021-2022). InSPACE-2 data provides the first insights into colloidal phase separation in toggled fields and unusual mechanical buckling instabilities. Our approach uses newly developed image analysis techniques to detect and track features in these datasets, allowing for more detailed physical interpretation than was previously possible. Over the past year, we have deepened understanding of self-assembly in toggled magnetic fields by focusing on magnetostatic energetics and emergent dynamics in the InSPACE data. Our advances include: 1. Understanding Surface Effects. We developed a way to calculate how the "outer layer" or surface of a particle cluster affects its shape and stability. Our findings show that, in the conditions we studied, surface forces are much stronger than random motion. This means the way the particles arrange themselves at the surface plays a large role in the shapes we see. 2. Understanding the Bulk of the Clusters. We also built a new method for studying what happens inside these clusters, not just at their surface. This method connects what's happening at the smallest scale – where individual particles line up in crystalline structures – to the larger shapes the clusters form. We found that the size and shape of a cluster can change the way magnetic forces work inside it, sometimes making it respond more strongly to a magnetic field than the individual particles would on their own. This could explain why scientists have seen unusually strong magnetic responses in some experiments. 3. Analysis of toggled-field data revealed a distinctive "ribbon" phase of extremely elongated aggregates coexisting with small dot-like clusters. Ribbons exhibit complex translational and rotational motion and continuously exchange smaller aggregates via two types of surface instabilities: one at ribbon ends, and one along ribbon ends. The end instability, in particular, involves the growth and detachment of small nodules, with ejection events recurring on timescales of roughly one minute. These observations highlight dynamic, far-from-equilibrium processes central to dissipative self-assembly.

	The project provided significant hands-on research experience to PhD student, who contributed to data analysis, model development, code creation for quantitative imaging, and manuscript preparation. He presented preliminary results at multiple scientific meetings and is expected to complete his dissertation by the end of 2025.
Bibliography Type:	Description: (Last Updated: 01/27/2026)
Abstracts for Journals and Proceedings	Furst EM. "Dissipative self-assembly and active matter behavior of magnetic colloidal suspensions in microgravity." 95th Annual Meeting of The Society of Rheology, Austin, TX, October 13-17, 2024. Abstracts. 95th Annual Meeting of The Society of Rheology, Austin, TX, October 13-17, 2024. , Oct-2024
Abstracts for Journals and Proceedings	Furst EM. "Dynamics of a colloidal suspension during dissipative self-assembly." ACS Fall 2025 Meeting (American Chemical Society), Washington, DC, August 17-21, 2025. Abstracts. ACS Fall 2025 Meeting (American Chemical Society), Washington, DC, August 17-21, 2025. , Aug-2025
Articles in Peer-reviewed Journals	Conradt J, Furst EM. "Magnetostatic energies in crystals of paramagnetic particles." Physical Review E. 2026 Jan 9;113(1):015407. https://doi.org/10.1103/pb3d-yg31 , Jan-2026
Articles in Peer-reviewed Journals	Conradt, J, Furst EM. "Quantitative imaging of colloidal structures." Langmuir. 2025 Mar 18;41(12):8176-91. https://doi.org/10.1021/acs.langmuir.4c05270 ; PubMed PMID: 40098481 , Mar-2025